
Algorithm 10: Bounded Forward Half-Search

Input:

- V . Set of nodes
- E_s . Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets (from Algorithm 3)
- $MTT(c)$. Precomputed minimum transit times (from Algorithm 5)
- $\text{bucket}(t)$. Time bucket mapping function
- MaxDuration . Maximum allowed elapsed time for the half-search (e.g., 840 mins / 14 hours)
- MinCountries . Minimum unique countries required to save the path (e.g., 6)
- GlobalMinTime . Minimum realistic time to visit a country (e.g., 25 mins)

Output:

- Labels_F . Map of nodes to a set of Pareto-optimal forward labels

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1: Initialize  $\text{Labels}_F(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize priority queue  $Q \leftarrow \emptyset$  // ordered by decreasing score
3: // Initialize labels at departures
4: for all  $v \in V$  do
5:   for all scheduled edges  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
6:      $L \leftarrow (\text{node} = u, t = \text{arr}, C = \{\text{country}(u)\}, \text{pred} = \text{null}, \text{start\_time} = \text{arr})$ 
7:      $\text{Labels}_F(u) \leftarrow \text{Labels}_F(u) \cup \{L\}$ 
8:      $\text{compute score}(L) \leftarrow (|L.C| \times 10000) - L.t$ 
9:      $Q \leftarrow Q \cup \{L\}$ 
10: // Main loop
11: while  $Q \neq \emptyset$  do
12:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  with maximum score from  $Q$ 
13:    $\text{elapsed} \leftarrow t - \text{start\_time}$ 
14:    $\text{remaining\_half\_time} \leftarrow \text{MaxDuration} - \text{elapsed}$ 
15:   // — 1. GLOBAL HARD FLOOR PRUNING (Half-Search Variant) —
16:    $\text{needed} \leftarrow \text{MinCountries} - |C|$ 
17:   // If the path physically cannot even reach the minimum required countries in the allowed 14 hours,
   kill it.
18:   if  $\text{needed} > 0 \wedge \text{remaining\_half\_time} < (\text{needed} \times \text{GlobalMinTime})$  then
19:     continue
20:   // — 2. TIME-BUDGETED UB PRUNING (Half-Search Variant) —
21:   if  $\text{needed} > 0$  then
22:      $b \leftarrow \text{bucket}(t)$ 
23:      $U \leftarrow R(v, b) \setminus C$ 
24:     Sort  $U$  as a sequence  $(c_1, c_2, \dots, c_k)$  such that  $MTT(c_1) \leq MTT(c_2) \leq \dots \leq MTT(c_k)$ 
25:      $\text{time\_spent} \leftarrow 0$ 
26:      $\text{UB} \leftarrow 0$ 
27:     for  $i = 1$  to  $|U|$  do
28:        $\text{time\_spent} \leftarrow \text{time\_spent} + MTT(c_i)$ 
29:       if  $\text{time\_spent} \leq \text{remaining\_half\_time}$  then
30:          $\text{UB} \leftarrow \text{UB} + 1$ 
31:       else

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32:   |   | break
33:   // If current countries + theoretically reachable countries  $j$  minimum acceptable, kill it.
34:   if  $|C| + \text{UB} < \text{MinCountries}$  then
35:   |   continue
36: // — 3. EXTEND VIA SCHEDULED EDGES —
37: for all  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
38:   if  $\text{dep} \geq t$  then
39:     new_elapsed  $\leftarrow \text{arr} - \text{start\_time}$ 
40:     // Enforce the 14-hour cut-off limit
41:     if new_elapsed  $\leq \text{MaxDuration}$  then
42:        $C' \leftarrow C \cup \{\text{country}(u)\}$ 
43:        $L' \leftarrow (u, \text{arr}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
44:       // Strict Pareto Dominance
45:       if  $\nexists L'' \in \text{Labels}_F(u)$  such that
46:         ( $L''.t \leq \text{arr} \wedge L''.C \supseteq C' \wedge (L''.t < \text{arr} \vee L''.C \neq C')$ ) then
47:           remove all  $L'' \in \text{Labels}_F(u)$  such that
48:             ( $\text{arr} \leq L''.t \wedge C' \supseteq L''.C \wedge (\text{arr} < L''.t \vee C' \neq L''.C)$ )
49:            $\text{Labels}_F(u) \leftarrow \text{Labels}_F(u) \cup \{L'\}$ 
50:           compute score( $L'$ )
51:            $Q \leftarrow Q \cup \{L'\}$ 
52: // — 4. EXTEND VIA TRANSFERS —
53: for all  $(v, w, \Delta) \in E_t$  do
54:    $t_{\text{new}} \leftarrow t + \Delta$ 
55:   new_elapsed  $\leftarrow t_{\text{new}} - \text{start\_time}$ 
56:   if new_elapsed  $\leq \text{MaxDuration}$  then
57:      $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
58:     // Strict Pareto Dominance
59:     if  $\nexists L'' \in \text{Labels}_F(w)$  such that
60:       ( $L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C)$ ) then
61:         remove all  $L'' \in \text{Labels}_F(w)$  such that
62:           ( $t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C)$ )
63:          $\text{Labels}_F(w) \leftarrow \text{Labels}_F(w) \cup \{L'\}$ 
64:         compute score( $L'$ )
65:          $Q \leftarrow Q \cup \{L'\}$ 
66: // — 5. CLEANUP —
67: // Purge any stored labels that survived Pareto dominance but failed to hit the final country threshold.
68: for all  $v \in V$  do
69:   for all  $L \in \text{Labels}_F(v)$  do
70:     if  $|L.C| < \text{MinCountries}$  then
71:       remove  $L$  from  $\text{Labels}_F(v)$ 
72: return  $\text{Labels}_F$ 

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